

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – II
PAPER –2
TEST DATE: 30-06-2022

ANSWERS, HINTS & SOLUTIONS

Physics

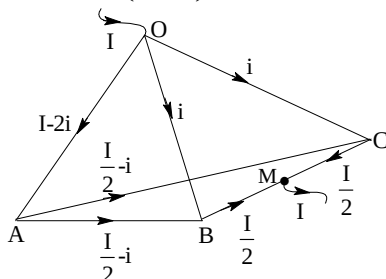
PART – I

Section – A

1. ABD

Sol. By Kirchoff's rule,

$$(1) ri = r\left(\frac{I}{2} - i\right) + r(I - 2i) \Rightarrow i = \frac{3}{8}I$$



$$(2) V = \frac{r}{2} \frac{I}{2} + ri = \frac{5}{8}rI$$

$$R_{eq} = \frac{V}{I} = \frac{5r}{8}$$

2. ABCD

Sol. Δx at O = d [path difference in maximum at O]

So if $d = 7\lambda / 2$, O will be a minima

$d = \lambda$, O will be maxima

$d = 5\lambda / 2$, O will be minima and hence intensity is minimum

$d = 4.8\lambda$ then total 10 minima can be observed on screen 5 above O and 5 below O, which corresponds to

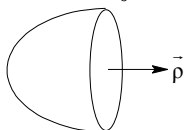
$$\Delta x = \pm \frac{\lambda}{2}, \pm \frac{3\lambda}{2}, \pm \frac{5\lambda}{2}, \pm \frac{7\lambda}{2}, \pm \frac{9\lambda}{2}$$

3. ABCD

4. BCD

Sol. (B) $U = -\rho E = -\frac{\sigma}{4\epsilon_0} \rho$

(C) $-\frac{\sigma\rho}{4\epsilon_0} + KE = 0 \Rightarrow KE = \frac{\sigma\rho}{4\epsilon_0}$



(D) $I\alpha = -\rho E \theta \Rightarrow \alpha = -\frac{\rho E}{I} \theta \Rightarrow \omega = \sqrt{\rho E / I}$

5. ACD

6. AB

7. A

8. D

Sol. (for Q. 7-8)

For the charge

$$\vec{F} = q(\vec{v} \times \vec{B}) - b\vec{v}$$

$$\therefore \frac{d\vec{v}}{dt} = \frac{q}{m} \left(\frac{d\vec{r}}{dt} \times \vec{B} \right) - \frac{b}{m} \left(\frac{d\vec{r}}{dt} \right)$$

$$\therefore \Delta\vec{v} = \frac{q}{m} (\Delta\vec{r} \times \vec{B}) - \frac{b(\Delta\vec{r})}{m}$$

9. D

10. C

Sol. Using IAOR

$$mg \frac{\ell}{2} \sin 53^\circ = mg \frac{\ell}{2} \sin 37^\circ + \frac{1}{2} I \omega^2$$

$$\omega = \sqrt{\frac{3}{5}} \Rightarrow V_{cm} = 5\sqrt{\frac{3}{5}} = \sqrt{15}$$

Section – B

11. 6

Sol. $E = \frac{V}{d} = \frac{200}{0.01} = 2 \times 10^4 \text{ V/m}$

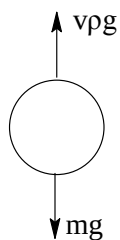
 When terminal velocity is achieved $qE = mg$

$$\Rightarrow n \times 1.6 \times 10^{-19} \times 2 \times 10^4 = \frac{4\pi}{3} (8 \times 10^{-7})^3 \times 900 \times 10 \Rightarrow n \approx 6$$

12. 4

Sol. $480 \times g = v \rho_1 g$

$$(480 - N)g = v \rho_2 g$$



$$\frac{480 - N}{480} = \frac{\rho_2}{\rho_1}$$

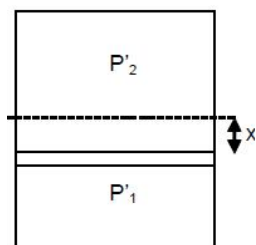
$$1 - \frac{N}{480} = 1 - \frac{50}{6000} \Rightarrow N = \frac{50 \times 480}{6000} = 4$$

13. 6

Sol. Assuming temperature remains constant at 300 K

$$\text{From } P_1 V_1 = P_2 V_2$$

$$\frac{P_1 \left(\frac{V_0}{2} \right)}{T} = \frac{P_1' \left(\frac{V_0}{2} - Ax \right)}{T}$$



$$3 \left(\frac{2x}{16 - x^2} \right) = 1$$

$$6x = 16 - x^2$$

$$x^2 + 6x - 16 = 0$$

$$x = 2$$

$$\text{Distance } 4 + 2 = 6$$

Section – C

14. 0.58

15. 0.75

16. 0.00

17. 4.00

18. 0.40

19. 1.20

Chemistry

PART – II

Section – A

20. A, B, C
 Sol. Neighboring group participation by a sulphur than involved.
21. C
 Sol. Process of active methylene group $\uparrow$ acidic strength and the side chain should be e^- withdrawing.
22. A, B, C
 Sol. Yield of 4-Hydroxypentan-2-one is low because $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ will form.
23. B
 Sol. Anode $\text{Ag} + \text{Br}^- \rightarrow \text{AgBr} + e^-$
 Cathode $\text{Ag}^+ + e^- \rightarrow \text{Ag}$

$$E_{\text{cell}}^0 = (E_{\text{RP}}^0)_C - (E_{\text{RP}}^0)_A$$

$$= E_{\text{cell}}^0 = -0.06 \log K_{\text{sp}}$$

$$Q = \frac{1}{[\text{Br}^-][\text{Ag}^+]} = \frac{1}{0.1 \times 0.1} = 100$$

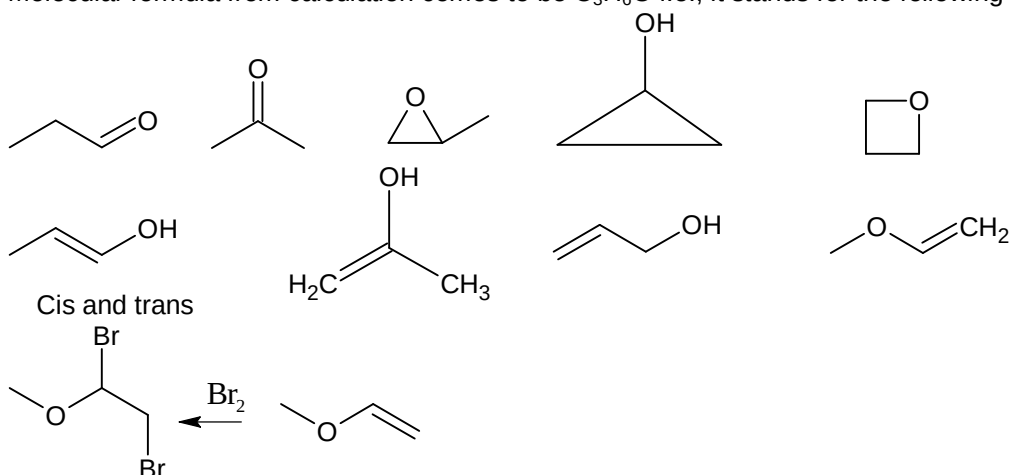
$$E_{\text{cell}}^0 = -0.06 \log K_{\text{sp}} - 0.06 \log 100$$

$$0.643 = -0.06 \log K_{\text{sp}} - 0.12$$

$$\log K_{\text{sp}} = \frac{0.763}{0.06} = -12.71$$
24. A, B, D
 Sol. Friedel Craft reaction not shown by Highly activated Rings
25. A, B, C
 Sol. DiBAL-H is electrophilic reducing agent as Borane and readily reduces electron rich species.
26. A
 Sol. AgCl is white precipitate and AgBr is pale yellow:
27. C
 Sol. No of unpaired e^- are 3.
28. B
29. B

Section – B

30. 1

Sol. Molecular formula from calculation comes to be C_3H_6O i.e., it stands for the following compounds.

$$VD \times 2 = MM \quad = 29 \times 2 = 58$$

$\therefore$ 58g of ether combines with $2 \times 80g Br_2$

$\therefore$ 2.9g of ether will combine with $\frac{2 \times 80}{58} \times 2.9 = 8g$ of Bromine

$Y=10$ and $X=8$

$$Y - X = 10 - 8 = 2$$

31. 8

Sol. $C_V = 3R/2$

AC polytropic $P \propto V$

$$\Rightarrow P \cdot V^{-1} = \text{constant}$$

$$C_{AC} = C_V + \frac{R}{1 - (-1)} = 2R$$

AB isobaric

$$C_{AB} = C_p = 5R/2$$

$$\therefore \frac{Q_{AC}}{Q_{AB}} = \frac{C_{AC}}{C_{AB}} [\because \Delta T \text{ same for both}]$$

$$= \frac{2R}{\left(\frac{5R}{2}\right)} = \frac{4}{5} = 0.8$$

32. 8

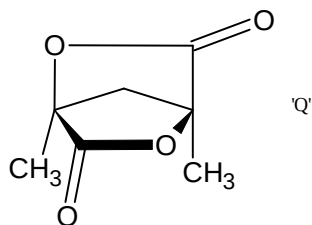
Sol. M mole of $Al_2(SO_4)_3 = 20$

Section – C

33. 4.00

34. 4.00

Sol.



35. 4.00

36. 1.00

Sol. (for Q. 35 – 36)

$$\text{Initial volume } V_0 = \frac{1}{3} L, \quad \text{final volume } V_f = \frac{4}{3} L$$

$$\text{Initial pressure } P_0 = 0.4 \text{ atm}, \quad \text{final pressure } P_f = 1.6 \text{ atm}$$

$$T_{\text{Initial}} = \frac{0.4}{3 \times nR}, \quad T_{\text{final}} = \frac{6.4}{3nR}$$

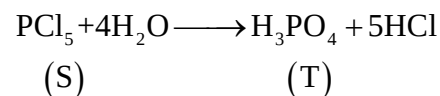
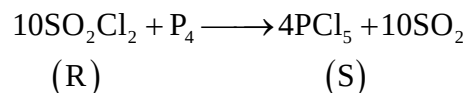
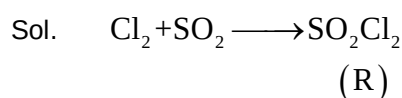
$$\Delta U = nC_{v,m}(T_f - T_0)$$

$$= n \times \frac{R}{\gamma - 1} \left[\frac{6}{3nR} \right] = n \times \frac{R}{1.5 - 1} \left[\frac{6}{3nR} \right] = 2nR \left[\frac{6}{3nR} \right] = 4$$

$$\begin{aligned} \text{Work by the gas} &= \int P_{\text{ext}} dv \\ &= \int 1.2V dv \\ &= \frac{1.2}{2} \left[(4V_0)^2 - V_0^2 \right] \\ &= \frac{1.2 \times 15V_0^2}{2} = \frac{1.2 \times 15}{2} \times \left(\frac{1}{3} \right)^2 = 1 \text{ litre atm} \end{aligned}$$

37. 3.00

38. 5.00



Mathematics**PART – III****Section – A**

39. A, B

40. B, D

Sol. Curve is $(y - 2)^2 = 2x^3 - 4$... (1)

$$\frac{dy}{dx} = \frac{3x^2}{y - 2}$$

∴ Equation of tangent at (x_1, y_1)

$$y - y_1 = \frac{3x_1^2}{y_1 - 2}(x - x_1)$$

As it passes through $(1, 2)$

$$\Rightarrow 2 - y_1 = \frac{3x_1^2}{y_1 - 2}(1 - x_1)$$

$$\text{or } (y_1 - 2)^2 = 3x_1^2(1 - x_1)$$

$$\text{or } 2x_1^3 - 4 = 3x_1^3 - 3x_1^2 \quad [\text{As } (x_1, y_1) \text{ lies on the curve}]$$

$$\text{or } x_1^3 - 3x_1^2 + 4 = 0$$

$$\text{or } (x_1 + 1)(x_1 - 2)^2 = 0$$

$$\therefore x_1 = -1 \text{ or } 2$$

But $x_1 \neq -1$ (it is clear from equation (1))

$$\therefore x_1 = 2$$

$$\therefore y_1 = 2 \pm 2\sqrt{3}$$

Points of contact are $(2, 2 + 2\sqrt{3})$ and $(2, 2 - 2\sqrt{3})$

∴ sum of x-coordinates of points of contact is 4

∴ sum of y-coordinates of points of contact is 4.

41. B, C

$$\text{Sol. If } \left| \int_a^b f(x) dx \right| = \int_a^b |f(x)| dx$$

⇒ $f(x)$ is either positive in (a, b) or $f(x)$ is negative in (a, b) and also $b > a$

$$\text{Also } g'(x) = f(x)$$

When $f(x) > 0 \Rightarrow g'(x) > 0 \Rightarrow g(x)$ is increasing functionWhen $f(x) < 0 \Rightarrow g'(x) < 0 \Rightarrow g(x)$ is decreasing functionalso $g(0) = 0$

$$\int_a^b f(x)g(x)dx = \int_a^b g(x)g'(x)dx$$

$$= \frac{(g(b))^2 - (g(a))^2}{2} \text{ which will be positive for } 0 < a < b.$$

42. A, C

 Sol. $C_1 \rightarrow C_1 - C_2$

$$\Rightarrow \Delta = \begin{vmatrix} 4 & (e^{i\alpha} - e^{-i\alpha})^2 & 4 \\ 4 & (e^{i\beta} - e^{-i\beta})^2 & 4 \\ 4 & (e^{i\gamma} - e^{-i\gamma})^2 & 4 \end{vmatrix} = 0 \quad (\because C_1 \text{ \& } C_3 \text{ are identical}).$$

43. A, C

 Sol. We have $AB = A$

$$\Rightarrow A(BA) = A [\because BA = B]$$

$$\Rightarrow (AB)A = A$$

$$\Rightarrow AA = A [\because AB = A]$$

$$\Rightarrow A^2 = A$$

 Again $BA = B$

$$\Rightarrow B(AB) = B [\because AB = A]$$

$$\Rightarrow (BA)B = B$$

$$\Rightarrow BB = B \Rightarrow B^2 = B$$

44. B, D

 Sol. $|z| - \log_2(z+2) = 0$

$$z+2 = 2^{|z|}$$

$$x+2 = 2^x \text{ and } y=0$$

 hence $x=2, y=0$.

45. A

46. B

47. A

48. D

Section – B

49. 0

50. 1

51. 3

Section – C

52. 768.00

53. 49.00

54. 44.00

55. 99.00

56. 3.00

57. 2.00